DATASHEET · REFERENCE SYSTEM

Smart Water Bottle Cap

Sensor cap for a Nalgene 32 oz Wide Mouth bottle: ESP32-C3 (Wi-Fi + BLE), VL53L4CD time-of-flight water level, Adafruit 381 DS18B20 temperature probe, LP5024 with 8 RGB LEDs, TPS63020 3V3, MCP73831 charger and MAX17048 fuel gauge on a 1000 mAh LP503450 cell, IP67 USB-C, in a printed PETG cap sealed with AS568 O-rings.



Assembled from UHD mates (3 placed bodies, 4 fasteners); strapped battery not shown.

BATTERY LIFE **89** d[S] minute samples, hourly sync	CELL **1000** mAh LP503450 LiPo	SLEEP CURRENT **53** µA[S] at the cell
LEVEL RANGE **18°** ToF field of view, 940 nm	PROBE ACCURACY **0.5** °C DS18B20, -10 to +85 °C	BOTTLE **946** mL Nalgene 32 oz Wide Mouth

ABOUT THIS DOCUMENT

Every value is queried from the UHD model at build time and re-checked by `uhd-tech-docs-verify`. Markers show where a value is not a cited source value.

[A] Assumption: the model marks this value as an assumption (assumption trait)

[D] Derived: computed from model values; formula in the provenance appendix

[S] Stand-in: stand-in data derived from the model, not measured

[–] Not in model: the model does not state this value (data gap)

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

00 The product

WHAT IT IS

A cap module that replaces the stock lid of a 32 oz Nalgene wide-mouth bottle and reports how much water is left and how warm it is, reminds the user to drink with a light ring, and syncs readings over Wi-Fi.

FEATURES

- Water level and temperature measured from the cap, with nothing in the bottle but a sealed stainless probe
- A ring of 8 RGB LEDs for reminders and status, and one button
- Wi-Fi sync, with provisioning over Bluetooth LE
- A 1000 mAh LiPo cell charged from USB-C through an IP67 receptacle; target of 30 days or more between charges
- A printed PETG cap that seals on the bottle's lip; hand wash only, no ingress rating claimed

WHAT IT DOES

ID	FUNCTION	DONE BY
F1	Measure the water level from the cap by time of flight to the surface	VL53L4CD looking down through a window
F2	Measure the temperature inside the bottle with a sealed stainless probe	DS18B20 probe through the cap floor
F3	Show reminders and status on a ring of RGB LEDs	LP5024 driving 8 RGB LEDs, light guide in the cap
F4	Sync readings to a server over Wi-Fi; provision over BLE	ESP32-C3-MINI-1 module
F5	Charge from USB-C; update firmware over USB or Wi-Fi	MCP73831, IP67 USB-C receptacle, ESP32-C3 USB Serial/JTAG
F6	Report battery state of charge	MAX17048
F7	One button: wake, show level, pairing	Side-push tact switch behind a sealed plunger

REQUIREMENTS

REQUIREMENT	HOW IT IS SHOWN
Bottle: Nalgene 32 oz Wide Mouth: 946 ml, Ø89 mm, 21 cm tall, 63 mm cap	*Not stated*
Battery: One LiPo cell, 1000 mAh, 5 × 34 × 50 mm	*Not stated*
Battery life: ≥ 30 days between charges with level sampling every minute and hourly sync	*Not stated*
Charge time: ≤ 5 h from empty	*Not stated*
Charging: USB-C 5 V through an IP67 receptacle, no USB Power Delivery	*Not stated*
Level range: Full bottle to empty: up to about 210 mm from the sensor window to the bottle floor	*Not stated*
Level resolution: ± 5 mm on a calm surface	*Not stated*
Temperature accuracy: ± 0.5 °C at the probe, -10 to +85 °C	*Not stated*
Cap envelope: ≤ 80 mm diameter (the bottle is 89 mm), ≤ 45 mm above the bottle's lip	*Not stated*
Cap weight: ≤ 120 g including the cell	*Not stated*
Ingress, bottle seal: Leak-proof when closed and inverted, at Nalgene's recommended closing torque	*Not stated*
Ingress, electronics: Survives hand washing and splashes with the USB-C port open; design intent equivalent to IPX7 with the lid sealed. No rating is claimed or tested	*Not stated*
Cleaning: Hand wash only; not dishwasher-safe	*Not stated*
Operating temperature: 0 °C to 40 °C ambient; water 0 °C to 60 °C at the probe	*Not stated*

From docs/brief.md. 14 of 14 do not say how they are shown.

01 Overview

SMART-BOTTLE · V0.1.0

Sensor cap for a Nalgene 32 oz Wide Mouth bottle: ESP32-C3 (Wi-Fi + BLE), VL53L4CD time-of-flight water level, Adafruit 381 DS18B20 temperature probe, LP5024 with 8 RGB LEDs, TPS63020 3V3, MCP73831 charger and MAX17048 fuel gauge on a 1000 mAh LP503450 cell, IP67 USB-C, in a printed PETG cap sealed with AS568 O-rings.

ROLE	MODULE	QTY
Cap housing	Bottle cap housing	1
Cap board	Bottle cap board	1
LiPo cell	LP503450 1000 mAh 3.7 V LiPo with PCM	1
Temperature probe	Adafruit 381 waterproof DS18B20 temperature probe	1
Bottle	Nalgene 32oz Wide Mouth Bottle, Tritan Renew	1
Lip seal O-ring	AS568-138 O-ring, silicone 70A, FDA grade	1
Lid seal O-ring	O-ring AS568-040, silicone 70A (72.75 x 1.78 mm)	1
Probe seal O-ring	AS568-010 O-ring, silicone 70A, FDA grade	1

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SYSTEM ARCHITECTURE

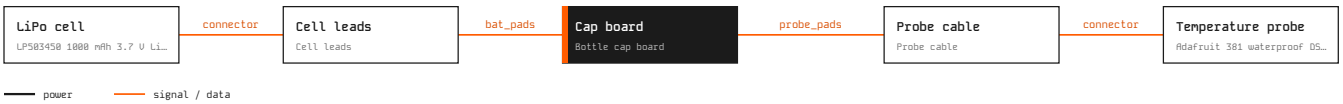


Figure 1 Power and signal links as stored in the model (validated with validateLinks); labels are the hub-side interfaces.

BOTTLE CAP BOARD ARCHITECTURE

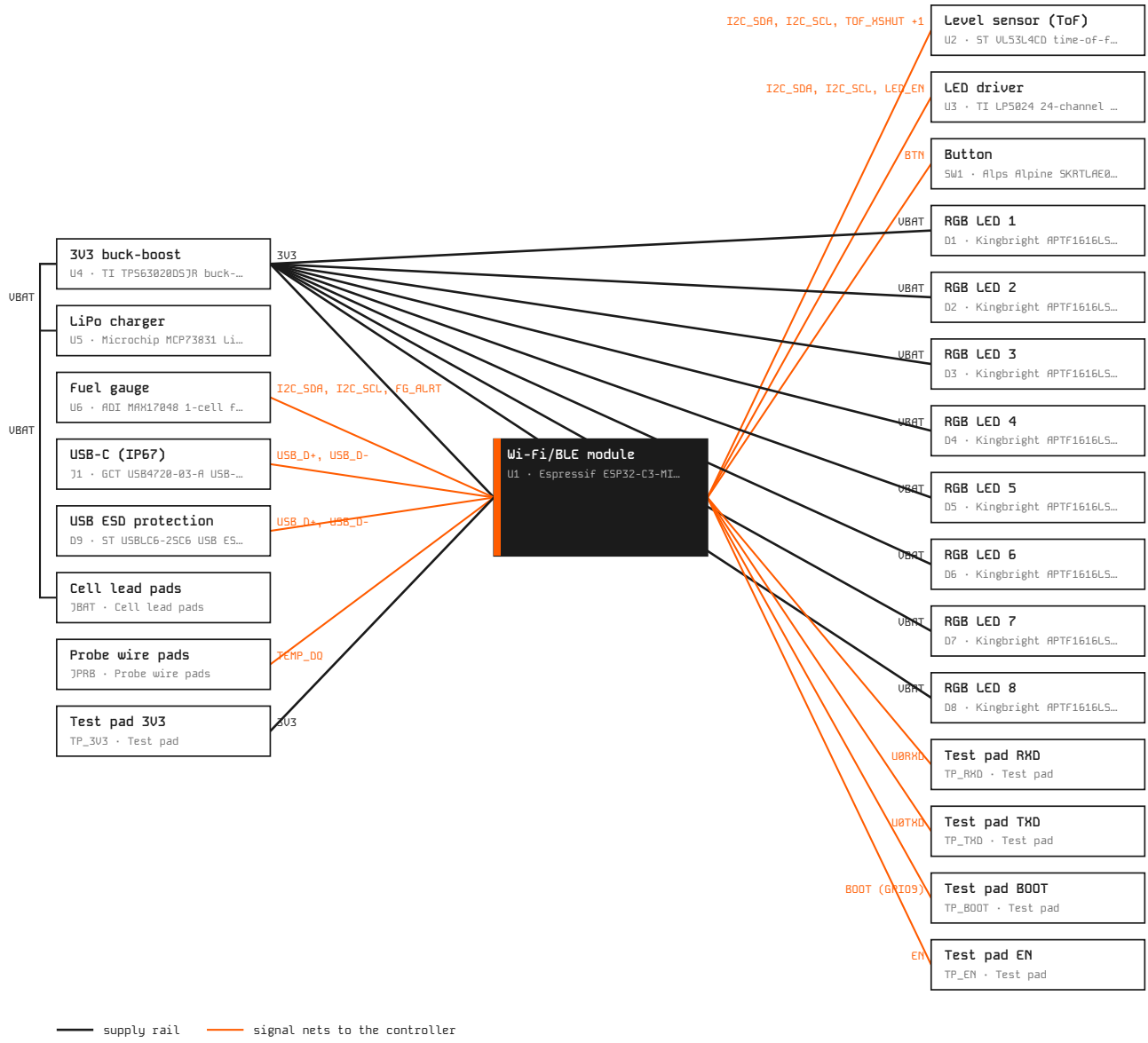


Figure 2 The board's components (passives left out) and the nets that join them to its controller, from the board module's net links. Parts with no net to the controller hang off the part they share a rail with.

02 Specifications

PARAMETER	VALUE
SENSING	
Level sensor	ST VL53L4CD time-of-flight sensor (VL53L4CDV0DH/1, Optical LGA12)
ToF I2C address	0x29
Temperature probe	Adafruit 381 waterproof DS18B20 temperature probe
Probe range	-55–125 °C
Probe conversion (12 bit)	750 ms
PROCESSING AND RADIO	
Module	Espressif ESP32-C3-MINI-1-H4X module (PCB antenna, 4 MB flash)
Supply	3.3 V (3–3.6 V)
Wi-Fi TX peak	350 mA
Deep-sleep current	5 µA
LIGHT RING	
LED driver	TI LP5024 24-channel I2C RGB LED driver (VQFN-32)
LEDs	Kingbright APTF1616LSEEZGKQBKC RGB LED (1.6 × 1.6 mm, common anode)
POWER	
Cell	LP503450 1000 mAh 3.7 V LiPo with PCM
Cell voltage	3.70 V
Charger	Microchip MCP73831 Li-ion charger, 4.20 V (2×3 DFN-8)
Charge current range	15–500 mA
3V3 converter	TI TPS63020DSJR buck-boost converter
Converter output rating	2 A
Fuel gauge	ADI MAX17048 1-cell fuel gauge (TDFN-8)
USB-C	GCT USB4720-03-A USB-C receptacle (USB 2.0, IP67, mid-mount)
Constant-current charge from empty, plus the taper	4.0 h[S]

PARAMETER	VALUE
MECHANICAL	
Cap	Bottle cap housing
Bottle	Nalgene 32oz Wide Mouth Bottle, Tritan Renew
Bottle size	88.9 × 88.9 × 209.55 mm
Neck finish	63 mm (63-415 per the Thermo lab-line torque table)
Closing torque (max)	6.33 N·m
Bottle mass	177.25 g

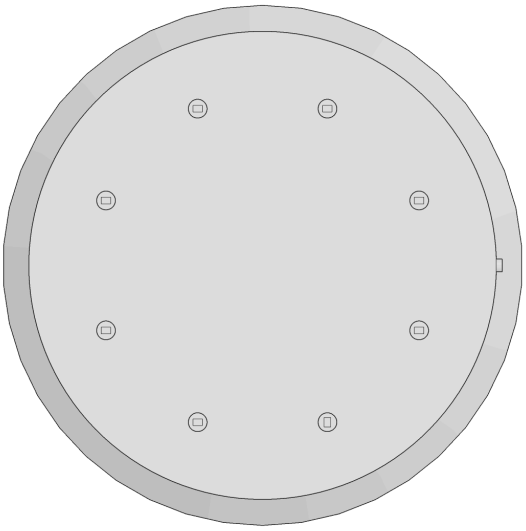
[A] Assumption: the model marks this value as an assumption (assumption trait)

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03 Mechanical



ENVELOPE

PARAMETER	VALUE
Cap envelope (max)	80 × 80 × 57 mm
Board envelope (max)	71 × 71 × 5 mm
Neck thread OD	62.07 mm
Neck thread pitch	4.230 mm
Board bolt circle	54 mm
Cap mass (printed)	73.7 g[A]
Cap material	PETG[A]

The cap's mass is the generated CAD volume × the declared material density (systemMass), so it follows the design. It is Ø80 mm overall: the bayonet lid needs no screw lugs (decisions D-B09).

Figure 3 Top view of the assembled cap on the bottle. (unit: mm)

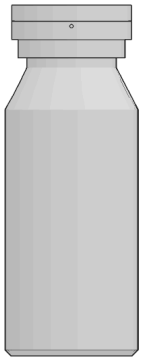


Figure 4 Right side: the cap on the Nalgene 32 oz Wide Mouth bottle. (unit: mm)

MASS BUDGET

PART	QTY	EACH	BASIS
Nalgene 32oz Wide Mouth Bottle, Tritan Renew	1	177.25 g	stated
Bottle cap housing	1	73.7 g[A]	CAD × PETG
LP503450 1000 mAh 3.7 V LiPo with PCM	1	20 g	stated
Bottle cap board	1	9 g	stated
AS568-138 O-ring, silicone 70A, FDA grade	1	1.3 g[A]	CAD × FDA silicone rubber, VMQ, 70 ± 5 Shore A
O-ring AS568-040, silicone 70A (72.75 × 1.78 mm)	1	0.83 g[A]	CAD × Silicone
M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	4	0.19 g	stated
AS568-010 O-ring, silicone 70A, FDA grade	1	0.08 g[A]	CAD × Silicone rubber

All-up weight 283.0 g[A] (systemMass), of which 75.9 g[A] is computed from assumed material or nominal CAD geometry. Not included, no mass in the model: Adafruit 381 waterproof DS18B20 temperature probe ×1, Cell leads ×1, Probe cable ×1. Rows in order of total mass (quantity × each).

INTERFACE GEOMETRY

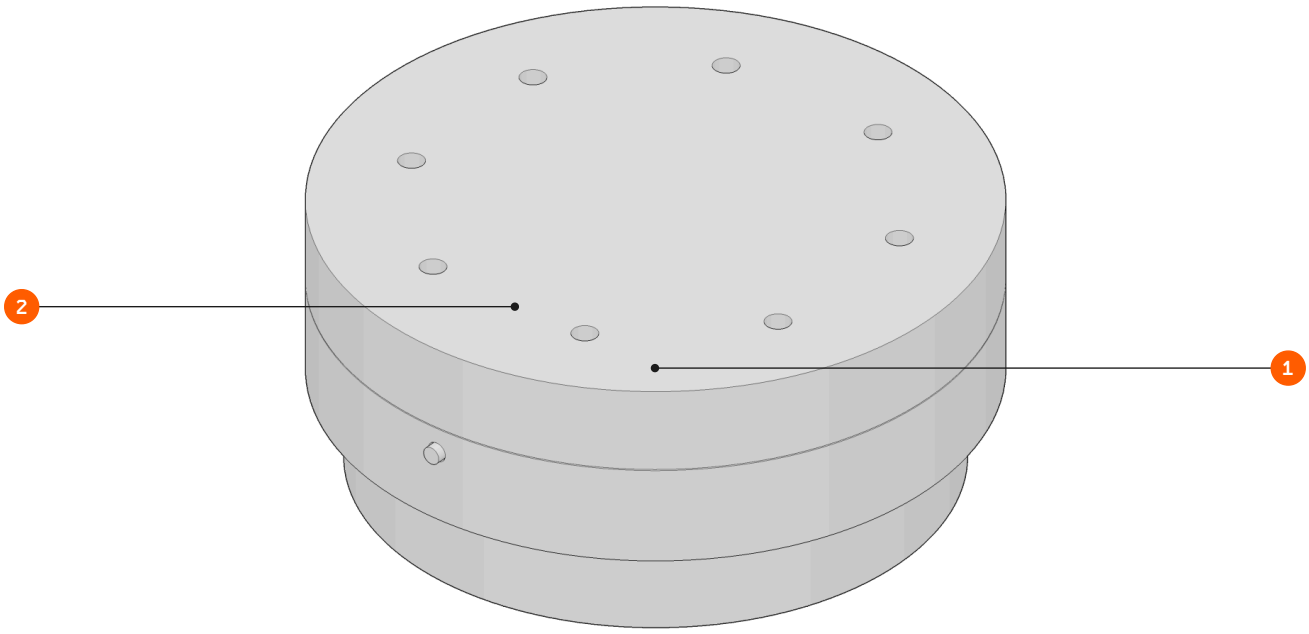


Figure 5 Structural interfaces of the frame and top plate (accent), from each interface's geometry frame and CAD feature binding.

#	INTERFACE	PATTERN	FASTENER	MATES WITH	BINDING
1	**Bottle lip seal (AS568-138)** cap.lip_seal	— × on —	—	Nalgene 32oz Wide Mouth Bottle, Tritan Renew	NONE
2	**Probe tube seal (AS568-010)** cap.probe_seal	— × on —	—		NONE

Binding: D1 named CAD feature, D2 per-interface artifact, D3 procedural from parameters (docs/geometry-artifacts.md).

04 Interfaces and pinout

Allocation of the Bottle cap board (`board`) interfaces, pad to pad, from the stored links. Crossovers (TX→RX) are marked ×. The link state is the result of pairwise DRC.

HUB INTERFACE	PADS (HUB → PEER)	PEER	PROTOCOL	PARAMETERS	STATE
Cell leads on the BAT pads	1 → P1 2 → P2	Cell leads Soldered end (BAT pads)	CONNECTOR	—	CONFIGURED
Probe cable on the probe pads	1 → P1 2 → P2 3 → P3	Probe cable Soldered end (probe pads)	CONNECTOR	—	CONFIGURED

Pin allocation of the Bottle cap board's controller, from the board module's nets: every controller pin on a net other than ground, with what else is on that net. Passives are listed by their role.

PAD	PIN	NET	ALSO ON THE NET
3	3V3	**3V3**	U4 VOUT (3V3 buck-boost) U2 AVDDVCSEL (Level sensor (ToF)) U2 AVDD (Level sensor (ToF)) JPRB VDD pad (red): 3V3 to the probe (Probe wire pads) TP_3V3 Test pad (Test pad 3V3) R11, C10, C11, C12, C1, C2, C4, C5, R1, R2, R3, R4, R6, R7, R8, R16
5	IO2	**I2C_SCL**	U2 SCL (Level sensor (ToF)) U3 SCL (LED driver) U6 SCL (Fuel gauge) R4
6	IO3	**BTN**	SW1 2 (Button) R8
8	EN	**EN**	TP_EN Test pad (Test pad EN) R1, C3
12	IO0	**TOF_XSHUT**	U2 XSHUT (Level sensor (ToF)) R5
13	IO1	**TOF_INT**	U2 GPIO1 (Level sensor (ToF)) R6
16	IO10	**VBUS_SENSE**	R17, R18
18	IO4	**TEMP_DQ**	JPRB DQ pad (yellow/white): 1-Wire data to the probe (Probe wire pads) R7
19	IO5	**LED_EN**	U3 EN (LED driver) R9
20	IO6	**FG_ALRT**	U6 ALRT (Fuel gauge) R16
21	IO7	**CHG_STAT**	R14, R15
22	IO8	**I2C_SDA**	U2 SDA (Level sensor (ToF)) U3 SDA (LED driver) U6 SDA (Fuel gauge) R3
23	IO9	**BOOT (GPIO9)**	TP_BOOT Test pad (Test pad BOOT) R2
26	IO18	**USB_D-**	J1 Dn1 (USB-C (IP67)) J1 Dn2 (USB-C (IP67)) D9 I/O2 (USB ESD protection)
27	IO19	**USB_D+**	J1 Dp1 (USB-C (IP67)) J1 Dp2 (USB-C (IP67)) D9 I/O1 (USB ESD protection)
30	RXD0	**U0RXD**	TP_RXD Test pad (Test pad RXD)
31	TXD0	**U0TXD**	TP_TXD Test pad (Test pad TXD)

05 Electrical characteristics

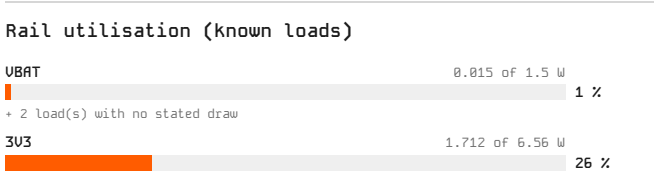


Figure 6 checkSystem supply_budget.

POWER RAILS

RAIL	VOLTAGE	CAPACITY	KNOWN LOAD	LOADS	UNBUDGETED
VBAT · Battery charge output (VBAT + VSS) board/u5:battery_charge	4.2 V[A]	1.5 W	0.0151 W	board/jbat:pad_pos, board/jbat:pad_pos, board/u4:vin, board/u6:supply, board/u3:supply	board/jbat:pad_pos, board/u4:vin
3V3 · Regulated output (VOUT) board/u4:vout	1.2–5.5 V	6.56 W	1.71 W	board/u1:pin_3, board/u1:pin_3, board/u2:supply	—

From checkSystem rule supply_budget: capacity = rail voltage × current rating; loads that state no draw cannot be budgeted. A branch rail (supplied_from) is budgeted on the output that feeds it; A marks a relation assumed, not cited.

SUPPLY INPUTS

MODULE	INPUT	VOLTAGE RANGE	MIN. SUPPLY POWER
Espressif ESP32-C3-MINI-1-H4X module (PCB antenna, 4 MB flash)	3V3 3	3.3 V[A]	—
ST VL53L4CD time-of-flight sensor (VL53L4CDV0DH/1, Optical LGA12)	AVDDVCSEL 1	2.8 V	—
ST VL53L4CD time-of-flight sensor (VL53L4CDV0DH/1, Optical LGA12)	AVDD 11	2.8 V	—
ST VL53L4CD time-of-flight sensor (VL53L4CDV0DH/1, Optical LGA12)	Supply (AVDD + AVDDVCSEL, GND + AVSSVCSEL)	2.8 V	—
TI LP5024 24-channel I2C RGB LED driver (VQFN-32)	VCC 27	2.7–5.5 V	—
TI LP5024 24-channel I2C RGB LED driver (VQFN-32)	Supply (VCC + GND pad)	2.7–5.5 V	—
TI TPS63020DSJR buck-boost converter	VINA 1	1.8–5.5 V	—
TI TPS63020DSJR buck-boost converter	VIN 10	1.8–5.5 V	—
TI TPS63020DSJR buck-boost converter	VIN 11	1.8–5.5 V	—
TI TPS63020DSJR buck-boost converter	Input supply (VIN + VINA)	3.6 V	—
Microchip MCP73831 Li-ion charger, 4.20 V (2x3 DFN-8)	VDD 1	5.0 V[A]	—
Microchip MCP73831 Li-ion charger, 4.20 V (2x3 DFN-8)	VDD 2	5.0 V[A]	—
Microchip MCP73831 Li-ion charger, 4.20 V (2x3 DFN-8)	Charger input (VDD + VSS)	5.0 V[A]	—
ADI MAX17048 1-cell fuel gauge (TDFN-8)	VDD 3	2.5–4.5 V[A]	—
ADI MAX17048 1-cell fuel gauge (TDFN-8)	Cell supply and sense (VDD + GND + EP)	2.5–4.5 V[A]	—
ST USBLC6-2SC6 USB ESD Protection (SOT23-6L)	VBUS 5	5.0 V[A]	—
ST USBLC6-2SC6 USB ESD Protection (SOT23-6L)	VBUS clamp (VBUS + GND)	5.0 V[A]	—
LP503450 1000 mAh 3.7 V LiPo with PCM	Lead + (red) +	3.7 V	—
LP503450 1000 mAh 3.7 V LiPo with PCM	Charge input (+ / -), 4.2 V CC/CV, 500 mA max	4.2 V	—
Adafruit 381 waterproof DS18B20 temperature probe	VDD wire (red), 3.0-5.5 V VDD	3.0–5.5 V	—

Stored links: 11; configured: 11. Diagnostics on the nominal build: info fastener_torque.

06 Performance

TEST DATA

> ⚠ **CAUTION** Charts and figures tagged STAND-IN · NOT MEASURED are derived from the model to show the expected shape. Replace them by dropping measured files into `docs/test-data/` (status `measured`); the next build swaps them in.

BATTERY-LIFE ESTIMATE BY USE PROFILE

STAND-IN · NOT MEASURED

#	PROFILE	DAILY CHARGE (MAH)	BATTERY LIFE (D)
P1	baseline: samples every 5 min, 4 syncs[S]	3.14[S]	271[S]
P2	samples every minute, 4 syncs[S]	6.95[S]	122[S]
P3	samples every 5 min, hourly sync[S]	5.78[S]	147[S]
P4	target: samples every minute, hourly sync[S]	9.6[S]	89[S]
P5	target with poor Wi-Fi (20 s per sync)[S]	22.28[S]	38[S]

Stand-in: days = usable capacity ÷ daily charge (sleep floor + samples + syncs + reminders), scripts/power-budget.ts; method and assumptions in docs/power-budget.md. Conditions: Capacity 1000 mAh; Usable fraction (assumed) 85 %; Sleep floor at the cell 53 µA.

Battery life by row STAND-IN · NOT MEASURED

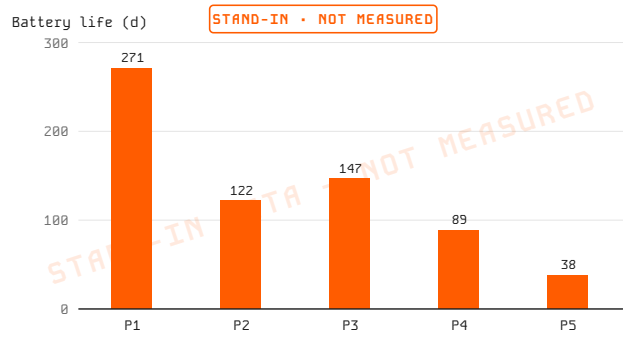


Figure 7 Battery-life estimate by use profile; P1-P5 are the table's rows.

PERFORMANCE SUMMARY

FIGURE	VALUE	TEST DATA
Target profile (minute samples, hourly sync)	89 d[S]	battery-life
Baseline (5 min samples, 4 syncs)	271 d[S]	battery-life
Sleep floor	53 µA[S]	battery-life
Charge from empty (constant current and taper)	4 h[S]	battery-life

STAND-IN ASSUMPTIONS

- usable capacity 85 % of rated; VBAT 3.7 V; buck-boost efficiency 90 % active, 70 % asleep
- sample: awake 0.3 s at 25 mA, 3 ranges of 0.05 s, a 0.75 s conversion
- sync: 4 s at 120 mA; reminder: 3 s, 3 mA per LED
- self-discharge of the cell not stated by the maker, not included

07 Design checks

CHECKSYSTEM

The nominal build and each fault scenario in `scenarios.ts` run through `checkSystem` (the system and each circuit board in it). A fault scenario is expected to raise at least one error or warning.

SCENARIO	DIAGNOSTICS	FINDINGS (ERRORS AND WARNINGS)
Nominal build nominal	0 ERR 0 WARN 3 INFO	no errors or warnings
ESP32-C3 module on USB VBUS module-on-vbus	1 ERR 0 WARN 3 INFO	NET board: u1:pin_3 (3V3) is rated 3–3.6 V but net vbus (VBUS) is 5 V.
Fuel gauge ground pad on VBAT gauge-gnd-on-vbat	2 ERR 0 WARN 3 INFO	LINK_STATE board: Battery charge output (VBAT + VSS) ↔ Cell supply and sense (VDD + GND + EP): incompatible — pin_6 is wired to ep, but power pairs it with pin_4 NET board: net vbat (VBAT) joins ground (u6:pin_4) to supply pins (jbat:pad_pos, u4:pin_10, u4:pin_11, u4:pin_1, u5:pin_3, u5:pin_4, u6:pin_3, u3:pin_27): a short.
Cell leads soldered the wrong way round cell-leads-reversed	2 ERR 0 WARN 3 INFO	LINK_STATE Lead + (red) ↔ BAT- pad (black lead): incompatible — no compatible protocol/role between cell:lead_pos and board:bat_neg LINK_STATE Lead - (black) ↔ BAT+ pad (red lead): incompatible — no compatible protocol/role between cell:lead_neg and board:bat_pos
Probe DQ and GND wires swapped probe-dq-on-gnd-pad	2 ERR 0 WARN 3 INFO	LINK_STATE DQ wire (yellow or white), 1-Wire data ↔ GND pad (black/blue): incompatible — no compatible protocol/role between probe:wire_dq and board:probe_gnd LINK_STATE GND wire (blue or black) ↔ DQ pad (yellow/white): 1-Wire data to the probe: incompatible — no compatible protocol/role between probe:wire_gnd and board:probe_dq
Cap bosses on a Ø50 circle bosses-on-50mm	1 ERR 0 WARN 3 INFO	LINK_STATE Board on the cap's insert bosses: incompatible — hole_spacing: 50 mm vs 54 mm (ranges do not overlap)

08 Bill of materials

DOCS/SMART-BOTTLE-BOM.JSON

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
1	Adafruit 381 waterproof DS18B20 temperature probe adafruit-381-ds18b20-probe	Adafruit Industries 381 (ADA381)	1	PURCHASE	—	not audited A 4 · gaps 5
2	ADI MAX17048 1-cell fuel gauge (TDFN-8) adi-max17048g-t10	Analog Devices (Maxim Integrated) MAX17048G+T10	1	PURCHASE	—	— A 2 · gaps 3
3	Alps Alpine SKRTLAE010 side-push TACT switch alps-skrtlae010	Alps Alpine SKRTLAE010	1	PURCHASE	—	not audited A 3 · gaps 3
4	AS568-010 O-ring, silicone 70A, FDA grade as568-010-silicone	Generic (SAE AS568) S70010 (The O-Ring Store); alternate Marco S1000-010	1	PURCHASE	0.08 g[A]	audited 17/17 A 1 · gaps 4
5	AS568-138 O-ring, silicone 70A, FDA grade as568-138-silicone	Generic (SAE AS568) S70138 (The O-Ring Store, AS568-138 S70 FDA)	1	PURCHASE	1.3 g[A]	audited 35/38 A 2 · gaps 5
6	Coilcraft XFL4020-152ME 1.5 µH Power Inductor 1.5 µH ±20 % coilcraft-xfl4020-152me	Coilcraft XFL4020-152ME	1	PURCHASE	0.169 g	audited 16/17 A 2 · gaps 4
7	Espressif ESP32-C3-MINI-1-H4X module (PCB antenna, 4 MB flash) espressif-esp32-c3-mini-1-h4x	Espressif Systems ESP32-C3-MINI-1-H4X	1	PURCHASE	—	audited 23/25 A 3 · gaps 5
8	GCT USB4720-03-A USB-C receptacle (USB 2.0, IP67, mid-mount) gct-usb4720-03-a	GCT USB4720-03-A	1	PURCHASE	—	audited 18/18 A 3 · gaps 5
9	Kingbright APTF1616LSEEZGKQBKC RGB LED (1.6 × 1.6 mm, common anode) kingbright-aptf1616lseezgkqbkc	Kingbright APTF1616LSEEZGKQBKC	8	PURCHASE	—	— A 1 · gaps 3
10	LP503450 1000 mAh 3.7 V LiPo with PCM lipol-lp503450	LiPoly Batteries Co., Ltd. (polybattery.com) LP503450	1	PURCHASE	20 g	audited 16/16 A 4 · gaps 6
11	M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless iso-7045-m2x6-a2	Generic (ISO 7045 / DIN 7985 Z) SPP-M2-6-A2	4	PURCHASE	0.19 g	not audited A 2 · gaps 4
12	Microchip MCP73831 Li-ion charger, 4.20 V (2×3 DFN-8) microchip-mcp73831t-2aci-mc	Microchip Technology MCP73831T-2ACI/MC	1	PURCHASE	—	not audited A 3 · gaps 5
13	Murata GRM155 MLCC (1005M / 0402, X5R / X7R) 1 µF ±10 % murata-grm155	Murata GRM155R61A105KE15D	3	PURCHASE	—	audited 10/10 A 2 · gaps 3
14	Murata GRM155 MLCC (1005M / 0402, X5R / X7R) 100 nF ±10 % murata-grm155	Murata GRM155R71C104KA88D	4	PURCHASE	—	audited 10/10 A 2 · gaps 3
15	Murata GRM188 MLCC (1608M / 0603, X5R / X7R) 10 µF ±20 % murata-grm188	Murata GRM188R61A106ME69D	4	PURCHASE	—	audited 8/8 A 2 · gaps 3
16	Murata GRM188 MLCC (1608M / 0603, X5R / X7R) 22 µF ±20 % murata-grm188	Murata GRM188R61A226ME15D	3	PURCHASE	—	audited 8/8 A 2 · gaps 3
17	Murata GRM188 MLCC (1608M / 0603, X5R / X7R) 10 µF ±20 % murata-grm188	Murata GRM188R61E106MA73D	1	PURCHASE	—	audited 8/8 A 2 · gaps 3
18	Nalgene 32oz Wide Mouth Bottle, Tritan Renew nalgene-32oz-wide-mouth	Nalgene 2020-3732 (Clear; most colours 2020-xx32, gray-black 682021-0333)	1	PURCHASE	177.25 g	not audited A 7 · gaps 5

Rows as written by the document build (docs/smart-bottle-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
19	O-ring AS568-040, silicone 70A (72.75 × 1.78 mm) as568-040-silicone	Generic (SAE AS568) S1000-040 (Marco Sealing Solutions, AS568-040 silicone 70A, FDA)	1	PURCHASE	0.83 g[A]	—– A 2 · gaps 2
20	ruthex RX-M2x4 brass heat-set insert, M2 × 4.0 mm ruthex-rx-m2x4	ruthex GE-M2x04-001	4	PURCHASE	0.16 g[A]	audited 21/21 A 1 · gaps 3
21	ST USBLC6-2SC6 USB ESD Protection (SOT23-6L) st-usblc6-2sc6	STMicroelectronics USBLC6-2SC6	1	PURCHASE	0.017 g	not audited A 2 · gaps 5
22	ST VL53L4CD time-of-flight sensor (VL53L4CDV0DH/1, Optical LGA12) st-vl53l4cdv0dh-1	STMicroelectronics VL53L4CDV0DH/1	1	PURCHASE	—	—– A 2 · gaps 3
23	TI LP5024 24-channel I2C RGB LED driver (VQFN-32) ti-lp5024rsm	Texas Instruments LP5024RSMR	1	PURCHASE	—	—– A 2 · gaps 3
24	TI TPS63020DSJR buck-boost converter ti-tps63020dsjr	Texas Instruments TPS63020DSJR	1	PURCHASE	—	audited 20/20 A 2 · gaps 2
25	Yageo RC0402 Chip Resistor (RC_L, 0402) 100 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-07100KL	3	PURCHASE	—	audited 9/9 A 2 · gaps 2
26	Yageo RC0402 Chip Resistor (RC_L, 0402) 10 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-0710KL	6	PURCHASE	—	audited 9/9 A 2 · gaps 2
27	Yageo RC0402 Chip Resistor (RC_L, 0402) 14.7 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-0714K7L	1	PURCHASE	—	audited 9/9 A 2 · gaps 2
28	Yageo RC0402 Chip Resistor (RC_L, 0402) 180 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-07180KL	1	PURCHASE	—	audited 9/9 A 2 · gaps 2
29	Yageo RC0402 Chip Resistor (RC_L, 0402) 1 M ohm ±1 % yageo-rc0402	Yageo RC0402FR-071ML	1	PURCHASE	—	audited 9/9 A 2 · gaps 2
30	Yageo RC0402 Chip Resistor (RC_L, 0402) 3.3 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-073K3L	1	PURCHASE	—	audited 9/9 A 2 · gaps 2
31	Yageo RC0402 Chip Resistor (RC_L, 0402) 4.7 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-074K7L	3	PURCHASE	—	audited 9/9 A 2 · gaps 2
32	Yageo RC0402 Chip Resistor (RC_L, 0402) 5.1 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-075K1L	2	PURCHASE	—	audited 9/9 A 2 · gaps 2
33	Yageo RC0402 Chip Resistor (RC_L, 0402) 68 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-0768KL	2	PURCHASE	—	audited 9/9 A 2 · gaps 2
34	Bottle cap board bottle-cap-board	—–	1	FABRICATE	9 g	—– A 3 · gaps 0
35	Bottle cap housing bottle-cap-enclosure	—–	1	FABRICATE	73.7 g[A]	—– A 0 · gaps 0
36	Cell lead pads (2 × Ø2.0 mm) bottle-battery-pads	—–	1	INCL. bottle-cap-board	—	—– A 1 · gaps 0
37	Cell leads bottle-battery-leads	LiPoly Batteries Co., Ltd.	1	INCL. lipol-lp503450	—	—– A 1 · gaps 0

Rows as written by the document build (docs/smart-bottle-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
38	Probe cable bottle-probe-cable	Adafruit Industries	1	INCL. adafruit-381-ds18b20-probe	—	— A 1 · gaps 0
39	Probe wire pads (3 × Ø1.8 mm) bottle-probe-pads	—	1	INCL. bottle-cap-board	—	— A 1 · gaps 0
40	Test pad (Ø1.2 mm) bottle-test-pad	—	6	INCL. bottle-cap-board	—	— A 1 · gaps 0

Rows as written by the document build (docs/smart-bottle-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

09 Evidence and assumptions

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Adafruit 381 waterproof DS18B20 temperature probe	5	not audited	A build variant: three-wire build: red VDD, blue or black GND, yellow or white DQ; no shield or drain wire A tube length: the 30 mm is the exposed stainless length, from the closed tip to the heat-shrink sleeve A generated heat-shrink geometry: sleeve behind the tube Ø6.3 × 24 mm, then a sleeve over the cable Ø4.8 × 25 mm A generated cable and wire-end geometry: only the rigid part is drawn: the stainless tube and the first 6 mm of the Ø6.3 heat shrin	manufacturer CAD; heat-shrink and wire-breakout dimensions; internal DS18B20 package and wire gauge; mass; cable length reference
ADI MAX17048 1-cell fuel gauge (TDFN-8)	6	not audited	A QSTRT voltage range: 0–5.5 V A CAD package dimensions: nominal body 2.00 × 2.00 × 0.75 mm, terminals 0.25 × 0.30 mm, exposed pad 1.20 × 0.80 mm w	—
Alps Alpine SKRTLAE010 side-push TACT switch	5	not audited	A typical wiring: terminal 2 to a GPIO input with a pull-up; terminals 1 and 3 to GND A terminal 1 position: terminal 1 at -X: the left terminal looking at the stem A CAD dimensions not on the drawing: terminal feet 0.15 mm thick, flat on the mounting face, 1.34–2.67 mm behind the frame face	manufacturer CAD; mass; terminal thickness and riser shape
AS568-010 O-ring, silicone 70A, FDA grade	7	17/17	A generated body: Nominal free-state torus, ID 6.07, CS 1.78 (centre-line radius 3.925, section radius 0.89)	manufacturer CAD; mass; SAE AS568 standard text; tensile strength (TORS VMQ-70)
AS568-138 O-ring, silicone 70A, FDA grade	7	35/38	A dimensional tolerance: AS568 limits: ID 53.64 ± 0.43 mm, CS 2.62 ± 0.08 mm A generated body: nominal torus: centre-line Ø56.26, section Ø2.62, lying on z = 0 (z 0..2.62); no flash, pa	manufacturer CAD; mass; pressure rating; tensile strength; silicone-specific dimensional tolerance
Coilcraft XFL4020-152ME 1.5 µH Power Inductor	7	16/17	A terminal designators: pin 1 = start (short) lead terminal under the marking bar; pin 2 = the other terminal A CAD package dimensions: nominal 4.0 × 4.0 body at the 2.10 maximum height; terminals 0.82 × 3.25 × 0.10 below the	manufacturer CAD; terminal designators; body corner radius, marking bar size; thermal resistance
Espressif ESP32-C3-MINI-1-H4X module (PCB antenna, 4 MB flash)	8	23/25	A 3V3 current_draw: 0.5 A on pin 3 A GPIO voltage parameter: 3.0–3.6 V on every logic pad and EN A CAD pads and antenna: pads at the STEP's centres and the drawing's sizes, 0.02 mm thick on the seating plane; an	manufacturer CAD; mass; module pad plating thickness; ADC2 oneshot reads on chip revision v1.1; land-pattern DXF
GCT USB4720-03-A USB-C receptacle (USB 2.0, IP67, mid-mount)	6	18/18	A D+/D- contacts on both sides: the board joins A6 (Dp1) with B6 (Dp2) and A7 (Dn1) with B7 (Dn2) A grounding-tag designator: TAG, one ground leaf for both tags A CAD body and internals: shell a stadium 8.94 × 3.26 mm (0.30 mm wall around the 8.34 opening), Z -1.13 to 2.13 fro	manufacturer CAD; mass; receptacle internals; lifecycle and stock; grounding tag designator

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Kingbright APTF1616LSEEZGKQBKC RGB LED (1.6 × 1.6 mm, common anode)	4	not audited	A CAD details not dimensioned: castellation dia 0.26 mm with its centre 0.054 mm outside the body edge; pad metal 0.01 mm	—
LP503450 1000 mAh 3.7 V LiPo with PCM	9	16/16	A charge and discharge temperature: charge 0 to +45 C, discharge -20 to +60 C A length includes the PCM: 50 mm overall, of which the last 5.0 mm (along the length) is the folded PCM zone at full A lead side and order: both leads leave the PCM end face next to the '+' side edge, red (+) outermost at y = +15. A lead wire and length: AWG28 UL1571, insulation Ø0.90 mm, conductor Ø0.38 mm, 50 mm beyond the body end; drawn as	manufacturer datasheet / drawing; manufacturer CAD; temperature ranges; PCM thresholds; lead gauge, length, strip length, position (+1)
M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	5	not audited	A ISO 7045 edition: M2 dimensions from ISO 7045:1994 (second edition) A CAD simplifications: plain Ø2.0 shank (thread major diameter, no thread form) with a flat end; under-head radiu	manufacturer CAD; Pozidriv recess form; tightening torque, proof load, operating temperature; as-rolled end form
Microchip MCP73831 Li-ion charger, 4.20 V (2×3 DFN-8)	3	not audited	A doubled VDD and VBAT pins: connect both VDD pins (1, 2) to the input and both VBAT pins (3, 4) to the cell A VBAT voltage lower bound: 0 V A CAD package dimensions: nominal A 0.90, A1 0.02, b 0.25, L 0.40; exposed pad 1.425 × 1.625 mm (midpoints of D2 1.3	manufacturer CAD; exposed pad nominal size; doubled VDD / VBAT pins; PROG logic-high threshold; VBAT minimum voltage
Murata GRM155 MLCC (1005M / 0402, X5R / X7R)	7	10/10	A terminal designators: pin_1 = the -X termination, pin_2 = the +X termination A CAD package dimensions: nominal 1.0 × 0.5 × 0.5 mm chip; each termination one block 0.25 mm long (middle of e 0.15	manufacturer CAD (none linked from the cited sources; one generated body stands for the series); mass (no unit mass in the cited sources); series lineup (which capacitance exists at which voltage and dielectric is not in the per-part reference sheets; a placement names its exact MPN)
Murata GRM188 MLCC (1608M / 0603, X5R / X7R)	6	8/8	A terminal designators: pin_1 = the -X termination, pin_2 = the +X termination A CAD package dimensions: nominal 1.6 × 0.8 × 0.8 mm chip; each termination one block 0.35 mm long (middle of e 0.2	manufacturer CAD (none linked from the cited sources; one generated body stands for the series); mass (no unit mass in the cited sources); series lineup (which capacitance exists at which voltage and dielectric is not in the per-part reference sheets; a placement names its exact MPN)

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Nalgene 32oz Wide Mouth Bottle, Tritan Renew	8	not audited	A neck finish of the consumer bottle: the SPI 63 mm finish (T, E, S, I, 6 TPI) and Nalgene's lab-line 63-415 closure facts A finish height H: 10.36 mm (SPI 400-series 63 mm, 9.98-10.74) A inner neck (bore) diameter: 53 mm A thread start, turns and profile: right-hand single start, 1.5 turns, lead-in on +X; upper flank 10° from radial, lower flan A lip, neck, shoulder and wall: flat land Ø55-Ø59.68, 1 × 45° inner chamfer; plain neck 8 mm below the finish; conical sho A height datum: 209.55 mm (8.25 in) from the base to the top of the lip, without the cap A closing torque: 40-56 in-lb (4.52-6.33 N·m)	manufacturer CAD; neck finish of the consumer bottle; inner neck diameter; minimum temperature; lip / neck / shoulder geometry
O-ring AS568-040, silicone 70A (72.75 × 1.78 mm)	5	not audited	A supplier and compound: Marco S1000-040 (FDA silicone 70A) as the representative SKU A CAD state: free (uncompressed, unstretched) at nominal ID and CS	density of compound S1000; shrinkage class of S1000; SAE AS568 document
ruthex RX-M2x4 brass heat-set insert, M2 × 4.0 mm	10	21/21	A generated body: 12-tooth knurl bands with a 4° flat crest and a V root at a constant twist (upper -50° ove	manufacturer CAD; mass; brass alloy, pull-out force, torque-out, tightening torque, operating temperature, installation temperature
ST USBLC6-2SC6 USB ESD Protection (SOT23-6L)	4	not audited	A VBUS working voltage: 0-5.25 V, nominal 5 V A CAD package dimensions: midpoint of each Table 3 range (body 2.9 × 1.6, span 2.8, lead 0.4 wide × 0.15 thick, foot	manufacturer CAD; datasheet revision; VBUS working voltage; internal I/O pin connection; thermal resistance
ST VL53L4CD time-of-flight sensor (VL53L4CDV0DH/1, Optical LGA12)	2	not audited	A CAD cap and apertures: cap 4.40 × 2.40 mm like the substrate, z 0.31-1.00; side recesses, 0.035 mm marks and draf A CAD pads: 0.02 mm thick, inside the substrate, faces on the seating plane	—
TI LP5024 24-channel I2C RGB LED driver (VQFN-32)	4	not audited	A package height (CAD, dimensions): 1.0 mm A CAD package dimensions: body 4.00 × 4.00 × 1.00 mm; terminals 0.20 × 0.35 mm flush with the body edge, centres 1.8	—
TI TPS63020DSJR buck-boost converter	5	20/20	A package height (CAD, dimensions): 0.9 mm A CAD lead size and pad thickness: 0.24 × 0.40 mm leads, 0.02 mm pads	manufacturer CAD; package height nominal
Yageo RC0402 Chip Resistor (RC_L, 0402)	1	9/9	A terminal designators: pin_1 = the -X termination, pin_2 = the +X termination A CAD package dimensions: nominal 1.00 × 0.50 × 0.35 mm chip; each termination one block 0.25 mm long (I2) over the	manufacturer CAD (none linked from the cited sources; one generated body stands for the series); mass (no unit mass in the cited sources)

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Bottle cap board	—	not audited	A capacitor MPNs GRM155R61A105KE15D (1 uF) and GRM188R61A226ME15D (22 uF): 1 uF 10 V X5R 0402 and 22 uF 10 V X5R 0603 A ToF supply capacitor: 10 uF on AVDDVCSEL (GRM188R61A106ME69D) and 0.1 uF on AVDD A mass: 9 g assembled	—
Bottle cap housing	—	not audited	—	—
Cell lead pads (2 × Ø2.0 mm)	—	not audited	A pad size: Ø2.0 mm, 3.5 mm apart	—
Cell leads	—	not audited	A gauge and insulation: AWG28 UL1571, Ø0.90 mm insulation	—
Probe cable	—	not audited	A drawn diameter: three Ø1.5 mm wires; the jacket (Ø4 mm) and heat shrink (Ø4.8-6.3 mm) are not drawn	—
Probe wire pads (3 × Ø1.8 mm)	—	not audited	A pad size: Ø1.8 mm, 2.6 mm apart	—
Test pad (Ø1.2 mm)	—	not audited	A pad size: Ø1.2 mm	—

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

10 Value provenance

This document shows 492 bound values. Each is a query into the UHD model, re-resolved and compared by `uhd-tech-docs-verify`; the full list with sources is in `uhd-bottle-ds.values.json` next to the document.

STATUS	VALUES	MEANING
Cited source	67	stated by a source in the part's sources.json
Model design value	344	a design value of a custom module, or a UHD check result
Derived	18	computed from model values; formula in the provenance appendix
Assumption	18	the model marks this value as an assumption (assumption trait)
Stand-in	22	stand-in data derived from the model, not measured
Not in model	23	the model does not state this value (data gap)

STATUS	VALUE	QUERY	DERIVATION / REASON
S Stand-in	89 d	test:battery-life:summary[id=target_days].value	docs/test-data/battery-life.json
S Stand-in	53 µA	test:battery-life:summary[id=sleep_floor_uA].value	docs/test-data/battery-life.json
S Stand-in	4.0 h	test:battery-life:summary[id=charge_h].value	docs/test-data/battery-life.json
A Assumption	73.7 g	derived:mass.module(bottle-cap-enclosure)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 58.04 cm³ × 1.27 g/cm³ (PETG); assumption: Prusament PETG's stated density stands in for the PETG the cap is printed in (cad.config.ts material); the m
A Assumption	PETG	def:bottle-cap-enclosure:domains[domain=mechanical].material.name	assumption trait · https://www.prusa3d.com/downloads/manuals/datasheets/prusament_petg_tds_en.pdf
A Assumption	1.3 g	derived:mass.module(as568-138-silicone)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.95 cm³ × 1.38 g/cm³ (FDA silicone rubber, VMQ, 70 ± 5 Shore A (TORS VMQ-70, legacy S70)); assumption: The specific gravity 1.38 is the supplier's com
A Assumption	0.83 g	derived:mass.module(as568-040-silicone)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.58 cm³ × 1.43 g/cm³ (Silicone (VMQ) 70 Shore A, Marco compound S1000 (density of Parker S0604-70 used)); assumption: Marco's S1000 datasheet gives no
A Assumption	0.08 g	derived:mass.module(as568-010-silicone)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.06 cm³ × 1.38 g/cm³ (Silicone rubber (VMQ), 70 Shore A, FDA 21 CFR 177.2600 grade (TORS VMQ-70)); assumption: The CAD body is the nominal free-state
A Assumption	283.0 g	derived:mass.all_up	systemMass: Σ quantity × (weight_g, else CAD volume × material density) over every instance; instances with neither are listed, not guessed · excludes 3 part(s) with no mass: Adafruit 381 waterproof DS18B20 temperature probe, Cell leads, Probe cable; 75.9 g of
A Assumption	75.9 g	derived:mass.assumed	Σ systemMass entries whose material carries an assumption · inputs: 4
– Not in model	–	inst:cap:interfaces[id=lip_seal].parameters[id=hole_count].value	
– Not in model	–	inst:cap:interfaces[id=lip_seal].parameters[id=hole_spacing].value	
– Not in model	–	inst:cap:interfaces[id=lip_seal].parameters[id=fastener_diameter].value	

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
– Not in model	—	inst:cap:interfaces[id=probe_seal].parameters[id=hole_count].value	
– Not in model	—	inst:cap:interfaces[id=probe_seal].parameters[id=hole_spacing].value	
– Not in model	—	inst:cap:interfaces[id=probe_seal].parameters[id=fastener_diameter].value	
A Assumption	4.2 V	def:microchip-mcp73831t-2aci-mc:interfaces[id=battery_charge].parameters[id=voltage].value	The data sheet gives no minimum VBAT: preconditioning applies below 66.5 % of VREG with no floor, so the pin follows the cell from 0 V up to VREG (4.232 V max). · https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP73
A Assumption	3.3 V	def:espressif-esp32-c3-mini-1-h4x:interfaces[id=pin_3].parameters[id=voltage].value	The GPIO are powered from VDD33 (chip Table 2-1) and their levels are fractions of it (Table 6-3); the parameter states the VDD33 range so a 3.3 V rail checks against it. · https://www.espressif.com/sites/default/files/documentation/esp32-c3-mini-1_datasheet_e
A Assumption	5.0 V	def:microchip-mcp73831t-2aci-mc:interfaces[id=pin_1].parameters[id=voltage].value	The data sheet gives no minimum VBAT: preconditioning applies below 66.5 % of VREG with no floor, so the pin follows the cell from 0 V up to VREG (4.232 V max). · https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP73
A Assumption	5.0 V	def:microchip-mcp73831t-2aci-mc:interfaces[id=pin_2].parameters[id=voltage].value	The data sheet gives no minimum VBAT: preconditioning applies below 66.5 % of VREG with no floor, so the pin follows the cell from 0 V up to VREG (4.232 V max). · https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP73
A Assumption	5.0 V	def:microchip-mcp73831t-2aci-mc:interfaces[id=supply].parameters[id=voltage].value	The data sheet gives no minimum VBAT: preconditioning applies below 66.5 % of VREG with no floor, so the pin follows the cell from 0 V up to VREG (4.232 V max). · https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP73
A Assumption	2.5–4.5 V	def:adi-max17048g-t10:interfaces[id=pin_3].parameters[id=voltage].value	The electrical characteristics give the -0.3 to +5.5 V data I/O range for SCL, SDA and ALRT only; QSTRT has just the -0.3 to +6 V absolute maximum. The data I/O range is used as the conservative operating limit. · https://www.analog.com/media/en/technical-docu
A Assumption	2.5–4.5 V	def:adi-max17048g-t10:interfaces[id=supply].parameters[id=voltage].value	The electrical characteristics give the -0.3 to +5.5 V data I/O range for SCL, SDA and ALRT only; QSTRT has just the -0.3 to +6 V absolute maximum. The data I/O range is used as the conservative operating limit. · https://www.analog.com/media/en/technical-docu
A Assumption	5.0 V	def:st-usblc6-2sc6:interfaces[id=pin_5].parameters[id=voltage].value	The datasheet states no recommended operating voltage. Leakage is specified at VRM = 5.25 V, which the Rev 4 change note ties to the USB standard, and VBR is 6 V min; 5.25 V is therefore the highest VBUS the stated figures cover. · https://static.chipdip.ru/li
A Assumption	5.0 V	def:st-usblc6-2sc6:interfaces[id=vbus_clamp].parameters[id=voltage].value	The datasheet states no recommended operating voltage. Leakage is specified at VRM = 5.25 V, which the Rev 4 change note ties to the USB standard, and VBR is 6 V min; 5.25 V is therefore the highest VBUS the stated figures cover. · https://static.chipdip.ru/li
S Stand-in	baseline: samples every 5 min, 4 syncs	test:battery-life:rows[0][0]	docs/test-data/battery-life.json

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
S Stand-in	3.14	test:battery-life:rows[0][1]	docs/test-data/battery-life.json
S Stand-in	271	test:battery-life:rows[0][2]	docs/test-data/battery-life.json
S Stand-in	samples every minute, 4 syncs	test:battery-life:rows[1][0]	docs/test-data/battery-life.json
S Stand-in	6.95	test:battery-life:rows[1][1]	docs/test-data/battery-life.json
S Stand-in	122	test:battery-life:rows[1][2]	docs/test-data/battery-life.json
S Stand-in	samples every 5 min, hourly sync	test:battery-life:rows[2][0]	docs/test-data/battery-life.json
S Stand-in	5.78	test:battery-life:rows[2][1]	docs/test-data/battery-life.json
S Stand-in	147	test:battery-life:rows[2][2]	docs/test-data/battery-life.json
S Stand-in	target: samples every minute, hourly sync	test:battery-life:rows[3][0]	docs/test-data/battery-life.json
S Stand-in	9.6	test:battery-life:rows[3][1]	docs/test-data/battery-life.json
S Stand-in	89	test:battery-life:rows[3][2]	docs/test-data/battery-life.json
S Stand-in	target with poor Wi-Fi (20 s per sync)	test:battery-life:rows[4][0]	docs/test-data/battery-life.json
S Stand-in	22.28	test:battery-life:rows[4][1]	docs/test-data/battery-life.json
S Stand-in	38	test:battery-life:rows[4][2]	docs/test-data/battery-life.json
S Stand-in	271 d	test:battery-life:summary[id=baseline_days].value	docs/test-data/battery-life.json
D Derived	0	derived:diagCount(nominal,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(nominal,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	3	derived:diagCount(nominal,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(module-on-vbus,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(module-on-vbus,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	3	derived:diagCount(module-on-vbus,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	2	derived:diagCount(gauge-gnd-on-vbat,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(gauge-gnd-on-vbat,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	3	derived:diagCount(gauge-gnd-on-vbat,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	2	derived:diagCount(cell-leads-reversed,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(cell-leads-reversed,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	3	derived:diagCount(cell-leads-reversed,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
D Derived	2	derived:diagCount(probe-dq-on-gnd-pad,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(probe-dq-on-gnd-pad,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	3	derived:diagCount(probe-dq-on-gnd-pad,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(bosses-on-50mm,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(bosses-on-50mm,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	3	derived:diagCount(bosses-on-50mm,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
– Not in model	–	sys:bom[line=2].verification	UHD bom
– Not in model	–	sys:bom[line=9].verification	UHD bom
– Not in model	–	sys:bom[line=19].verification	UHD bom
A Assumption	0.16 g	derived:mass.module(ruthex-rx-m2x4)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.02 cm³ × 8.5 g/cm³ (Brass (lead-free, cadmium-free; alloy not stated)); assumption: ruthex names no alloy; 8.5 g/cm³ is within the table's rolled-and
– Not in model	–	sys:bom[line=22].verification	UHD bom
– Not in model	–	sys:bom[line=23].verification	UHD bom
– Not in model	–	sys:bom[line=34].manufacturer	UHD bom
– Not in model	–	sys:bom[line=34].verification	UHD bom
– Not in model	–	sys:bom[line=35].manufacturer	UHD bom
– Not in model	–	sys:bom[line=35].verification	UHD bom
– Not in model	–	sys:bom[line=36].manufacturer	UHD bom
– Not in model	–	sys:bom[line=36].verification	UHD bom
– Not in model	–	sys:bom[line=37].verification	UHD bom
– Not in model	–	sys:bom[line=38].verification	UHD bom
– Not in model	–	sys:bom[line=39].manufacturer	UHD bom
– Not in model	–	sys:bom[line=39].verification	UHD bom
– Not in model	–	sys:bom[line=40].manufacturer	UHD bom
– Not in model	–	sys:bom[line=40].verification	UHD bom